CLAPS Bayesian Ordinal Workflow

Rationale, Pilot Results and Design Analysis

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2026-06-12

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1 Part I: Rationale

This part sets out the rationale for the CLAPS design analysis. It motivates the affectedness account behind the focal hypotheses, explains why the acceptability ratings are modelled with a Bayesian cumulative-ordinal mixed-effects model, gives the prior specifications that make the Bayes factors well defined and describes the simulation framework behind the power analysis.

1.1 Background and Theoretical Motivation

The CLAPS question is whether acceptability of passive and related sentence types tracks semantic affectedness, and whether this slope differs by sentence type. *Affectedness* — the degree to which a patient or theme participant undergoes a change of state or is causally impinged upon by the action — is a core dimension of argument structure (Beavers, 2011; Dowty, 1991), and it has long been argued to be the semantic core of the passive, which is associated with the meaning that the surface subject "is in a state or circumstance characterized by [the by-object] having acted upon it" (Pinker et al., 1987). Across languages, the prediction that passive acceptability is semantically structured rather than purely syntax-driven is supported by adult grammaticality-judgment and comprehension studies (Ambridge et al., 2016; Ambridge et al., 2023; Aryawibawa & Ambridge, 2018; Bidgood et al., 2020; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021).

Developmental and processing evidence converges on the same constraint: not all verb classes support passive comprehension or production equally — children acquire actional, highly affected-patient passives earlier and more reliably than non-actional ones — and affectedness-like semantic factors remain explanatory even when syntactic form is controlled (Agostinho et al., 2025; Darmasetiyawan et al., 2022; Maratsos et al., 1985; Nguyen & Pearl, 2021; Paolazzi et al., 2022; Pinker et al., 1987).

To date this cross-linguistic evidence base spans English, Indonesian, Balinese, Hebrew and Mandarin; the affectedness constraint on passive acceptability has not previously been tested in Turkish or Norwegian. CLAPS therefore extends the paradigm to two typologically and morphologically distinct languages for which no language-specific effect-size estimates yet exist — one reason the focal-slope priors are not anchored to language-specific values (see below).

This motivates a design analysis that quantifies whether realistic sample sizes, given ARC compute constraints, can recover the focal directional effects as Bayes factors under plausible data-generating values.

1.2 The Bayesian Cumulative-Ordinal Mixed-Effects Model

The CLAPS outcome is a discrete 1–7 Likert rating of sentence acceptability. Treating Likert responses as continuous and modelling them with a Gaussian likelihood biases regression coefficients and inflates both Type I and Type II error in the presence of floor or ceiling effects (Liddell & Kruschke, 2018). The cumulative-logit model is the principled alternative because it respects the discreteness, ordering and bounded nature of Likert data while leaving the latent scale free to vary continuously across conditions (Bürkner & Vuorre, 2019). Recent empirical psycholinguistic work has independently confirmed substantial inferential gains when Likert acceptability ratings are modelled cumulatively rather than metrically (Taylor et al., 2023; Veríssimo, 2021).

A Bayesian multilevel framework is required because Bayes factors depend on proper priors and full model likelihoods, uncertainty in hierarchical variance and correlation terms must be propagated, and the design analysis must evaluate directional evidence thresholds directly. The implementation uses `brms` / `cmdstanr` (Bürkner, 2017, 2018), with directional Savage–Dickey Bayes factors as the primary inferential quantity (Schad et al., 2023; Wagenmakers et al., 2010).

1.3 Empirical Basis for the Focal-Slope Prior Scales

The focal predictor is `Semantics_scaled`: a standardised measure of the semantic affectedness of the patient. The directional CLAPS hypothesis is that affectedness raises acceptability in the canonical passive and that this slope is attenuated, abolished or reversed in the active and in the pseudo-passive comparison sentence types. The central finding of the cross-language synthesis is exactly this pattern: affectedness predicts acceptability for both passives and actives but the effect is reliably *larger* for passives, and it is absent for pseudo-passives (Ambridge et al., 2023).

The previously observed magnitude of the affectedness slope in canonical passives, taken from independent samples of English, Indonesian, Balinese, Hebrew and Mandarin speakers, is of the order of 0.3 to 0.8 in the units reported by those studies (Ambridge et al., 2016; Aryawibawa & Ambridge, 2018; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021), with a cross-language pooled posterior mean of approximately 0.47 from the Bayesian meta-analytic synthesis (Ambridge et al., 2023). The S_Type $\times$ Semantics interaction (active minus passive) is correspondingly negative, of the order of -0.2 to -0.4. Because those antecedent studies modelled rating responses on their original (5- or 7-point) scales rather than on a cumulative-logit latent scale, these magnitudes are used as order-of-magnitude anchors for the log-odds-scale priors below, not as exact transcriptions; the prior scales are deliberately wide enough to absorb the resulting scale uncertainty.

The `primary` prior set therefore uses zero-centred Gaussians concentrated enough to cover the previously observed range without pre-loading the directional Bayes factor test. The Semantics slope receives `normal(0, 0.5)`, which places approximately 95% of prior mass between -1 and $+1$. The active–passive interaction receives the same scale. The pseudo-passive interaction receives a slightly wider `normal(0, 0.6)`: although the synthesis implies a negative pseudo-minus-passive interaction (affectedness does not raise pseudo-passive acceptability), pseudo-passive constructions are attested in only a subset of the contributing languages, so this interaction is less precisely constrained cross-linguistically than the active–passive one (Ambridge et al., 2023). The direction-encoding `literature_centred` sensitivity regime makes these anchors explicit, centring the three focal slopes on approximately 0.47, -0.31 and -0.36 respectively (see Appendix A); it is reported as a sensitivity-only check because it encodes the predicted direction.

1.4 Proper Priors for Bayes Factors

Marginal likelihoods, and therefore Bayes factors, are scale-sensitive functionals of the prior. Improper or extremely diffuse priors yield either undefined or arbitrarily small marginal likelihoods regardless of the data (Gelman et al., 2017; Gronau et al., 2017; Schad et al., 2023). All four prior regimes in `R/03_define_priors.R` are therefore proper. The Savage–Dickey ratio is computed under the primary regime; bridge sampling under all four regimes provides calibration (Gronau et al., 2017; Gronau et al., 2020).

The four-regime sensitivity grid follows Schad et al. (2023): the `primary` regime is zero-centred and weakly-to-moderately informative; `weak` relaxes all focal-slope scales; `literature_centred` is sensitivity-only and direction-encoding; `heavy_tailed` substitutes Student-t for Gaussian focal-slope priors.

1.5 Variance-Component and Correlation Priors

Hierarchical variance priors must be chosen carefully because Inverse-Gamma and other classical defaults concentrate prior mass away from zero and bias variance estimates upwards in small samples (Gelman, 2006). Chung et al. (2015) and Simpson et al. (2017) recommend half-Normal or half-Student-t priors on the standard-deviation scale. The CLAPS workflow therefore uses `student_t(3, 0, 1)` on all `sd` parameters across regimes.

Group-level correlation matrices use the LKJ prior (Lewandowski et al., 2009) with shape $\eta = 2$ in the primary, literature-centred and heavy-tailed regimes, which mildly concentrates prior mass toward the identity matrix and so regularises the many correlation parameters of the maximal random-effects structure; the `weak` regime uses $\eta = 1$, which is uniform over correlation matrices, as a sensitivity comparison.

1.6 Random-Effects Structure

The maximal random-effects ladder is motivated by Barr et al. (2013). Where models with fully crossed by-Participant and by-Verb random slopes fail to converge after the prespecified diagnostic budget, the workflow steps down one rung at a time following Matuschek et al. (2017). Treatment coding of `S_Type` with `Passive` as the reference level preserves direct interpretability of the focal coefficients (Brehm & Alday, 2022). Sample-size adequacy and the risk of optimistic generalisation from a single pilot are addressed using the framework of Westfall & Yarkoni (2016) and Albers & Lakens (2018).

1.7 Bayes-Factor Thresholds

CLAPS reports the conventional descriptive grades for Bayes factors ("anecdotal", "moderate", "strong", "very strong", "extreme"), which originate with Jeffreys (1961) and follow the modern relabelling of Lee & Wagenmakers (2013). These labels are descriptive only: directional Savage–Dickey ratios (Wagenmakers et al., 2010) on each focal coefficient are reported with full posterior quantification. Calibration via bridge sampling (Gronau et al., 2017; Gronau et al., 2020) is reported alongside the primary ratios.

1.8 Convergence Diagnostics and Prior-Predictive Checks

Every confirmatory fit must clear a fixed battery of diagnostics before it is counted: a rank-normalised, folded $\hat{R}$ below 1.01 on all parameters (Vehtari et al., 2021), bulk and tail ESS of at least 400 on all focal coefficients, zero post-warmup divergent transitions with no maximum-treedepth saturations (Gabry et al., 2019; Nicenboim et al., 2025) and a passed prior-predictive check confirming that the prior over Likert response distributions is plausibly varied without being degenerate (Gabry et al., 2019; Schad et al., 2021). Fits that fail any of these are stepped down the random-effects ladder rather than reported.

1.9 Pilot–Confirmatory Data Separation

Pilot data inform `compute_ceiling_calibrated_thresholds()` and `literature_centred` priors, but `split_pilot_confirmatory()` in `R/01_read_validate_data.R` ensures they are never passed to the confirmatory model. This separation is required for the validity of the sensitivity-only `literature_centred` regime, which would otherwise double-count pilot information (Albers & Lakens, 2018; Schad et al., 2023).

1.10 Simulation Design and Power Criteria

This is a Bayes-factor design analysis (BFDA): for each design cell it estimates, by simulation, the sampling distribution of the focal Bayes factors and hence the probability of obtaining compelling evidence at a given sample size (Schönbrodt & Wagenmakers, 2018; Stefan et al., 2019). Because the focal priors are anchored to previously observed effect magnitudes, the informed-prior BFDA variant of Stefan et al. (2019) is the appropriate template. Reporting the full operating characteristics, rather than a single power figure, also guards against the sign (Type S) and magnitude (Type M) errors that threaten inferences from small or noisy designs (Gelman & Carlin, 2014).

Each design *condition* combines language and design settings, model level, prior regime and sample size. To estimate a genuine power (exceedance) probability rather than a single-draw indicator, every condition is replicated across `n_simulations_per_cell` independently seeded simulated data sets — 200 for the single-language and gender conditions and 50 for the (8–12 h/fit) cross-language conditions — built by `scripts/generate_design_grid.R`, with one SLURM array task per replicate. Replicate fits use a lighter sampler (4 chains $\times$ 3000 iterations, `adapt_delta = 0.99`) sufficient for a single reliable Bayes factor; the heavy 16-chain $\times$ 5000-iteration sampler is reserved for the one-off maximal-model convergence demonstration that establishes the maximal feasible model and runtimes. SLURM arrays run replicates in parallel on ARC CPUs. For each condition, the analysis summarises the probability that Bayes factors exceed the primary and secondary thresholds, the sensitivity of the BF category to prior regime, the feasibility diagnostics and ladder-selection frequency and the convergence and sampler diagnostics. The descriptive evidence grades follow Jeffreys (1961) and Lee & Wagenmakers (2013), while interpretation remains tied to continuous BF values and robustness checks (Schad et al., 2023).

2 Part II: Pilot Results (Calibration Only)

This part reports the pilot-data analyses, which serve calibration purposes only. The pilot data are excluded from the final confirmatory analysis; the findings here inform prior choice and model-ladder decisions but do not constitute confirmatory evidence.

2.1 Pilot Data Overview

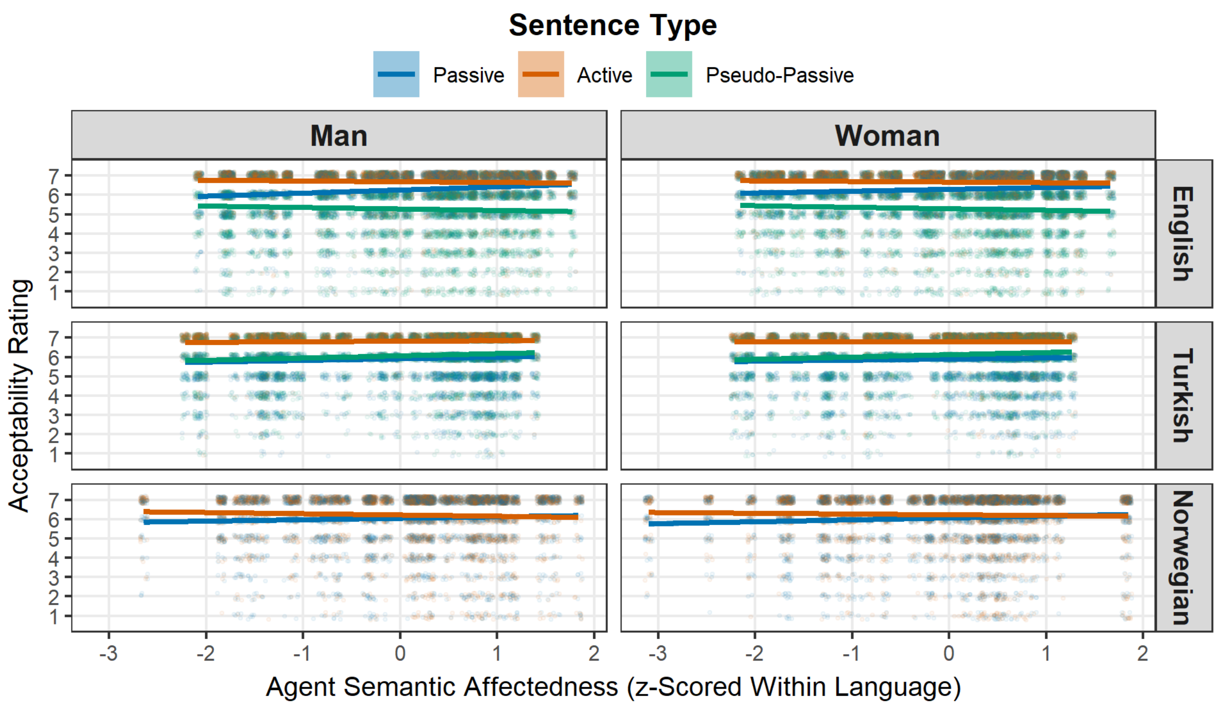
The pilot sample provides acceptability ratings for the three CLAPS languages, English, Norwegian and Turkish, across the passive, active and (where the construction is attested) pseudo-passive sentence types, with the agent-affectedness predictor measured for each item. These data are used only to calibrate the threshold priors and to choose the random-effects structure carried into the design analysis; they enter no confirmatory model.

2.2 Referent Gender

A further model variation adds referent gender as a covariate, to control for any unbalanced real-world association between particular actions and gender. Figure 1 shows the raw pilot acceptability data on the key comparison of sentence type against affectedness, split by gender and language. The sentence-type orderings and the affectedness slopes are closely matched for men and women within each language, so the focal effects are not being driven by a gender imbalance. This is consistent with the modelling result that adding the covariate barely shifts the power curve, so the gender model serves as a robustness check rather than a source of divergent conclusions.

Figure 1

Raw Pilot Acceptability by Agent Semantic Affectedness, Sentence Type, Referent Gender and Language



Note. Points are individual ratings (jittered). Lines show per-sentence-type linear trends. The Norwegian pilot has no pseudo-passive. Pilot data are used for calibration only.

2.3 Model Ladder Results

The model ladder was traversed from L5 (correlated maximal) downward, with the selected level for each language being the highest that compiled, converged and showed no material posterior pathologies. In the pilot fits the full correlated-maximal structure (L5) was estimable for each single language, so that level was carried into the design analysis. The pooled cross-language model required one step down to an uncorrelated maximal structure (L4), as the design-analysis feasibility results in Part III confirm.

2.4 Convergence Diagnostics

Convergence was assessed using $\hat{R}$ (Vehtari et al., 2021), bulk and tail ESS, the number of divergent transitions and maximum tree-depth exceedances. Every fit retained for calibration met the publication-grade thresholds set out in Part I; fits that did not were stepped down the ladder rather than retained.

2.5 Prior Sensitivity Results

The sensitivity analysis evaluates how Bayes-factor estimates and posterior summaries change across the four prior regimes (primary, weak, literature-centred and heavy-tailed) and the two threshold modes (broad and ceiling-calibrated). The quantitative four-regime comparison, computed on the design-analysis simulations, is reported in Part III under Prior Sensitivity of BF Category. At the pilot stage the comparison served only to confirm that prior choice does not change the qualitative conclusions.

2.6 Threshold Prior Calibration

For Turkish, the pilot data showed strong upper-end clustering. Ceiling-calibrated threshold priors were derived using smoothed cumulative category proportions and `qlogis()` (Bürkner & Vuorre, 2019; Gelman, 2006; Gelman et al., 2017). Prior predictive simulations confirmed that ceiling-calibrated priors produce distributions more consistent with the Turkish pilot response profile.

2.7 Conclusions for Prior Determination

The pilot analysis fixes the inputs to the design analysis. The `primary` prior is retained as prespecified, with `ceiling_calibrated` thresholds for Turkish, where the pilot ratings clustered at the upper end. The maximal feasible model is the correlated-maximal structure (L5) for each single language and the uncorrelated cross-language structure (L4) for the pooled model. Prior choice does not alter the focal conclusions, as the four-regime comparison in Part III confirms.

3 Part III: Design Analysis Results

This part reports the design analysis itself: the simulation framework and focal hypotheses, the feasibility of the planned models, the Bayes-factor power curves across sample size, their sensitivity to the prior and the resulting per-language sample-size recommendation.

3.1 Design Analysis Framework

The Bayes-factor design analysis estimates, via simulation, the probability that the prespecified Bayes-factor thresholds are exceeded for each focal hypothesis under a range of sample sizes, prior regimes and model-ladder levels (Schönbrodt & Wagenmakers, 2018; Stefan et al., 2019).

The focal hypotheses are directional. H1a states that the affectedness slope is positive for passives, $\beta_{\text{Semantics}} > 0$. H1b states that this slope is smaller for actives than for passives, $\beta_{\text{S\_TypeActive:Semantics}} < 0$, and it carries the main theoretical weight. In languages with a pseudo-passive, a secondary hypothesis (H2) concerns $\beta_{\text{S\_TypePseudo\_Passive:Semantics}}$, whose direction is reported both ways.

The primary success metric is $BF_{10} > 10$ (strong evidence, after Jeffreys (1961)) and the secondary metric is $BF_{10} > 3$ (moderate evidence). Bayes factors are computed via the Savage–Dickey directional density ratio from `brms` models with `sample_prior = "yes"` (Schad et al., 2023; Wagenmakers et al., 2010), with bridge sampling as a calibration check only (Gronau et al., 2017; Gronau et al., 2020).

3.2 Failure and Feasibility Summary

Every design cell that was launched produced a usable fit. The summary below records the successes against the possible failure modes (estimation error, timeout, out-of-memory and malformed output).

Table 1

Failure and Feasibility Summary Across All Design Cells

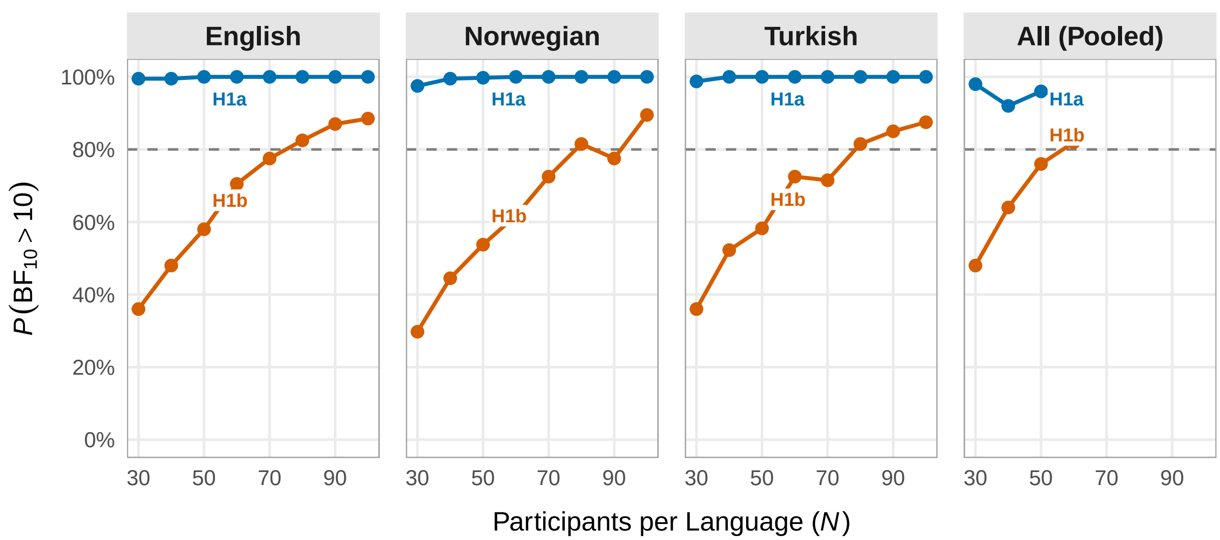
total_cells	n_success	n_error	n_timeout	n_oom	n_malformed	failure_rate
30800	30800	0	0	0	0	0

3.3 BF Exceedance Probabilities by Sample Size

The probability of exceeding the Bayes-factor thresholds rises with sample size for every focal hypothesis. The figure shows the power curves for the two focal hypotheses under the primary prior. The full operating characteristics across all conditions, prior regimes and hypotheses are tabulated in Appendix D.

Figure 2

Bayesian Power Curves for the Two Focal Hypotheses, by Language



Note. Points are exceedance probabilities, $P(BF_{10} > 10)$, under the primary prior. The dashed line marks the conventional 80% target. The pooled cross-language analysis was simulated to $N = 60$.

3.4 Prior Sensitivity of BF Category

Prior choice shifts the Bayes-factor power only modestly. The table reports the power for each focal hypothesis under all four prior regimes at the anchor sample size, where every regime was simulated. The semantic effect (H1a) is at ceiling throughout, while the interaction (H1b) moves within a narrow band.

Table 2

Prior Sensitivity of Statistical Power, as a Percentage, Across Prior Regimes at N = 50 (Maximal Model, 20 Verbs)

Language	Hypothesis	primary	heavy_tailed	literature_centred	weak
English	H1a	100	100	100	100
English	H1b	58	61	61	66
Norwegian	H1a	100	100	100	100
Norwegian	H1b	54	52	41	66
Turkish	H1a	100	100	100	100
Turkish	H1b	58	55	62	72

3.5 Model Ladder Selection Frequency

The ladder rarely needed to step down. The table records how often each level was the highest feasible model across the simulated fits.

Table 3

Frequency With Which Each Model-Ladder Level Was the Highest Feasible Model

model_level	n_selected	prop_selected
L4_cross_uncorrelated	800	0.025974
L5_correlated_maximal	30000	0.974026

3.6 Maximal Feasible Model

The maximal feasible model is the highest-complexity ladder level that both fit and met the publication-grade convergence criteria set out in Part I. The table gives it for each language alongside the number of converged cells.

Table 4

Maximal Feasible (Converged) Model Level per Language

language	maximal_feasible_model	level_rank	n_converged_cells
AllLanguages	L4_cross_uncorrelated	4	452
English	L5_correlated_maximal	5	11944
Norwegian	L5_correlated_maximal	5	5968
Turkish	L5_correlated_maximal	5	11940

3.7 Fit Runtimes

Fit cost varied widely by model and language, with the pooled cross-language fits much heavier than the single-language ones. The table summarises per-cell runtimes and convergence rates.

Table 5

Per-Cell Fit Runtimes (Minutes) and Convergence Rate, by Language and Model Level

language	model_level	n_cells	median_runtime_min	max_runtime_min	p_converged
AllLanguages	L4_cross_uncorrelated	800	239.50	593.96	0.56
English	L5_correlated_maximal	12000	17.09	64.79	1.00
Norwegian	L5_correlated_maximal	6000	8.68	31.55	0.99
Turkish	L5_correlated_maximal	12000	17.16	66.85	1.00

3.8 Conclusions and Recommended Sample Size

The design analysis yields a per-language sample-size recommendation at the assumed effect, together with the maximal model that can be estimated for each.

For English, the maximal feasible model is `L5_correlated_maximal`, and the recommended sample size is 80 participants (20 verbs) to reach P(BF > 10) of at least 80% on both focal hypotheses under the primary prior. For Norwegian, the maximal feasible model is `L5_correlated_maximal`, and the recommended sample size is 80 participants (20 verbs) to reach P(BF > 10) of at least 80% on both focal hypotheses under the primary prior. For Turkish, the maximal feasible model is `L5_correlated_maximal`, and the recommended sample size is 80 participants (20 verbs) to reach P(BF > 10) of at least 80% on both focal hypotheses under the primary prior. For the pooled cross-language analysis, the maximal feasible model is `L4_cross_uncorrelated`, and the recommended sample size is 60 participants (20 verbs) to reach P(BF > 10) of at least 80% on both focal hypotheses under the primary prior.

Across converged fits the median cell took 17.1 minutes, with the heaviest reaching 594.0 minutes.

Each exceedance probability above is a Monte-Carlo estimate over `n_simulations_per_cell` independently seeded replicates — 200 for single-language and gender conditions (Monte-Carlo $SE \approx 0.028$ near a true rate of 0.8) and 50 for cross-language conditions — so the recommended sample sizes are stable long-run rates rather than the indicative 0/1 values of the earlier single-seed feasibility run (archived under `outputs/design_analysis_feasibility_v1`, which also documents the L5 maximal-model convergence and per-fit runtimes). All recommendations remain conditional on the data-generating parameters encoded in `scripts/generate_design_grid.R`, whose sensitivity is examined via the prior regimes in the full exceedance table.

One assumption requires particular care. These curves condition on assumed true effect sizes anchored to the published affectedness estimates, and that anchor warrants caution. The relevant literature is subject to publication bias and the structural pressures of publish-or-perish culture, which together tend to inflate reported effects. Studies reporting positive findings, especially small and low-powered ones, systematically overstate the magnitude of what they detect, the winner's curse or Type M error (Button et al., 2013; Gelman & Carlin, 2014; Ioannidis, 2008). Design analyses built on the surviving published estimates then tend to be optimistic about power and replicability (Albers & Lakens, 2018; Vasishth et al., 2018). The implication is one-directional: if the true interaction is smaller than the literature-anchored value, the sample sizes here are a lower bound and the real requirement is larger. Because power depends steeply on effect size, that gap may be substantial. As an approximate guide, a true effect around two-thirds of the assumed magnitude roughly doubles the N needed for the same power, which would move the requirement from $N = 80$ towards $N = 160$ or beyond. $N = 80$ is therefore best treated as a floor rather than a target, and where feasible the design should plan against a deliberately conservative ('safeguard') effect size rather than the literature point estimate (Perugini et al., 2014).

Appendices

The appendices reproduce the prespecified prior definitions and the model-ladder formulas exactly as they are defined in the analysis scripts. Each begins on a new page.

Appendix A

Prior Specifications

The following empirical anchors are sourced from R/03_define_priors.R:

Anchor	Value
`semantics_pooled`	0.47
`s_type_active_interaction`	-0.31
`s_type_pseudo_passive_interaction`	-0.36
`semantics_min`	0.27
`semantics_max`	0.8

Prior regimes are sourced from R/03_define_priors.R, where N denotes a normal prior and t a Student-t prior:

Regime	default	semantics	active	pseudo	Intercept	sd	cor
primary	N(0, 1.5)	N(0, 0.5)	N(0, 0.5)	N(0, 0.6)	t(3, 0, 2.5)	t(3, 0, 1)	lkj(2)
weak	N(0, 2)	N(0, 1)	N(0, 1)	N(0, 1)	t(3, 0, 2.5)	t(3, 0, 2)	lkj(1)
literature_centred	N(0, 1.5)	N(0.47, 0.35)	N(-0.31, 0.4)	N(-0.36, 0.5)	t(3, 0, 2.5)	t(3, 0, 1)	lkj(2)
heavy_tailed	t(3, 0, 1.5)	t(3, 0, 0.5)	t(3, 0, 0.5)	t(3, 0, 0.6)	t(3, 0, 2.5)	t(3, 0, 1)	lkj(2)

Appendix B
Single-Language Model Ladder

The following ladder definitions are sourced from R/04_model_formulas.R via build_model_ladder(has_pseudo_passive = TRUE).

L5_correlated_maximal

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type * Semantics_scaled |
    Participant) + (1 + S_Type | Verb)
```

L4_uncorrelated_maximal

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type * Semantics_scaled ||
    Participant) + (1 + S_Type || Verb)
```

L3_no_participant_interaction_slope

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type + Semantics_scaled ||
    Participant) + (1 + S_Type || Verb)
```

L2_sentence_type_slopes_only

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type || Participant) +
    (1 + S_Type || Verb)
```

L1_random_intercepts_plus_participant_semantics

```
Response ~ S_Type * Semantics_scaled + (1 + Semantics_scaled ||
    Participant) + (1 | Verb)
```

L0_random_intercepts_only

```
Response ~ S_Type * Semantics_scaled + (1 | Participant) + (1 |
    Verb)
```

Appendix C
Cross-Language Model Ladder

The following ladder definitions are sourced from R/04_model_formulas.R via build_multilanguage_ladder().

L5_cross_maximal

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type * Semantics_scaled |
    Participant) + (1 + S_Type * Semantics_scaled | Verb) + (1 +
    S_Type * Semantics_scaled | Language)
```

L4_cross_uncorrelated

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type * Semantics_scaled ||
    Participant) + (1 + S_Type * Semantics_scaled || Verb) +
    (1 + S_Type * Semantics_scaled || Language)
```

L3_cross_no_participant_interaction

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type + Semantics_scaled ||
    Participant) + (1 + S_Type * Semantics_scaled || Verb) +
    (1 + S_Type * Semantics_scaled || Language)
```

L2_cross_stype_participant_verb

```
Response ~ S_Type * Semantics_scaled + (1 + S_Type || Participant) +
    (1 + S_Type || Verb) + (1 + S_Type * Semantics_scaled ||
    Language)
```

L1_cross_intercepts_only_ppt_verb

```
Response ~ S_Type * Semantics_scaled + (1 | Participant) + (1 |
    Verb) + (1 + S_Type || Language)
```

L0_cross_intercepts_only

```
Response ~ S_Type * Semantics_scaled + (1 | Participant) + (1 |
    Verb) + (1 | Language)
```

Appendix D

Full BF Operating Characteristics

Table 6

Probability of Exceeding BF > 10 (Primary) and BF > 3 (Secondary) per Design Condition

N	Verbs	Prior	Hyp	P(BF>10)	P(BF>3)	Median BF	Sims
30	20	primary	H1a	0.98	1.00	28	50
30	20	primary	H1b	0.48	0.96	10	50
30	20	primary	H2a	0.08	0.70	5	50
30	20	primary	H2b	0.00	0.02	0	50
40	20	primary	H1a	0.92	1.00	30	50
40	20	primary	H1b	0.64	0.90	12	50
40	20	primary	H2a	0.30	0.70	6	50
40	20	primary	H2b	0.00	0.00	0	50
50	20	primary	H1a	0.96	1.00	26	50
50	20	primary	H1b	0.76	0.98	13	50
50	20	primary	H2a	0.26	0.76	5	50
50	20	primary	H2b	0.00	0.00	0	50
60	20	primary	H1a	0.94	1.00	32	50
60	20	primary	H1b	0.82	1.00	20	50
60	20	primary	H2a	0.24	0.66	5	50
60	20	primary	H2b	0.00	0.00	0	50
30	20	primary	H1a	1.00	1.00	1369	400
30	20	primary	H1b	0.36	0.69	6	400
30	20	primary	H2a	0.34	0.69	5	400
30	20	primary	H2b	0.00	0.03	0	400
40	20	primary	H1a	1.00	1.00	5970	400
40	20	primary	H1b	0.48	0.78	9	400
40	20	primary	H2a	0.41	0.74	8	400
40	20	primary	H2b	0.00	0.01	0	400
50	20	heavy-tailed	H1a	1.00	1.00	984372	200
50	20	heavy-tailed	H1b	0.61	0.84	17	200
50	20	heavy-tailed	H2a	0.40	0.66	6	200
50	20	heavy-tailed	H2b	0.00	0.03	0	200
50	20	lit-centred	H1a	1.00	1.00	97996	200
50	20	lit-centred	H1b	0.61	0.84	16	200
50	20	lit-centred	H2a	0.54	0.82	11	200
50	20	lit-centred	H2b	0.00	0.00	0	200
50	20	primary	H1a	1.00	1.00	986343	400
50	20	primary	H1b	0.58	0.80	13	400
50	20	primary	H2a	0.42	0.74	8	400
50	20	primary	H2b	0.00	0.02	0	400
50	20	weak	H1a	1.00	1.00	987083	200
50	20	weak	H1b	0.66	0.88	28	200
50	20	weak	H2a	0.30	0.58	4	200
50	20	weak	H2b	0.02	0.06	0	200
60	20	primary	H1a	1.00	1.00	994017	400
60	20	primary	H1b	0.70	0.92	24	400
60	20	primary	H2a	0.46	0.74	8	400
60	20	primary	H2b	0.00	0.01	0	400
70	20	primary	H1a	1.00	1.00	996754	200
70	20	primary	H1b	0.78	0.94	43	200
70	20	primary	H2a	0.43	0.76	8	200
70	20	primary	H2b	0.00	0.02	0	200
80	20	primary	H1a	1.00	1.00	995509	200
80	20	primary	H1b	0.82	0.94	79	200

Table 6

Probability of Exceeding BF > 10 (Primary) and BF > 3 (Secondary) per Design Condition (continued)

N	Verbs	Prior	Hyp	P(BF>10)	P(BF>3)	Median BF	Sims
80	20	primary	H2a	0.56	0.82	12	200
80	20	primary	H2b	0.00	0.00	0	200
90	20	primary	H1a	1.00	1.00	995758	200
90	20	primary	H1b	0.87	0.98	74	200
90	20	primary	H2a	0.58	0.80	13	200
90	20	primary	H2b	0.00	0.02	0	200
100	20	primary	H1a	1.00	1.00	998001	200
100	20	primary	H1b	0.88	0.96	133	200
100	20	primary	H2a	0.52	0.78	11	200
100	20	primary	H2b	0.00	0.03	0	200
30	20	primary	H1a	0.98	1.00	402	400
30	20	primary	H1b	0.30	0.67	5	400
40	20	primary	H1a	1.00	1.00	3948	400
40	20	primary	H1b	0.44	0.76	8	400
50	20	heavy-tailed	H1a	1.00	1.00	951717	200
50	20	heavy-tailed	H1b	0.52	0.83	11	200
50	20	lit-centred	H1a	1.00	1.00	97243	200
50	20	lit-centred	H1b	0.41	0.70	7	200
50	20	primary	H1a	1.00	1.00	962227	400
50	20	primary	H1b	0.54	0.81	11	400
50	20	weak	H1a	1.00	1.00	982405	200
50	20	weak	H1b	0.66	0.90	21	200
60	20	primary	H1a	1.00	1.00	986590	400
60	20	primary	H1b	0.62	0.88	15	400
70	20	primary	H1a	1.00	1.00	998500	200
70	20	primary	H1b	0.72	0.92	31	200
80	20	primary	H1a	1.00	1.00	999499	200
80	20	primary	H1b	0.81	0.94	43	200
90	20	primary	H1a	1.00	1.00	996007	200
90	20	primary	H1b	0.78	0.95	57	200
100	20	primary	H1a	1.00	1.00	995260	200
100	20	primary	H1b	0.90	0.98	132	200
30	20	primary	H1a	0.99	1.00	1464	400
30	20	primary	H1b	0.36	0.71	6	400
30	20	primary	H2a	0.41	0.74	7	400
30	20	primary	H2b	0.00	0.00	0	400
40	20	primary	H1a	1.00	1.00	8085	400
40	20	primary	H1b	0.52	0.80	11	400
40	20	primary	H2a	0.40	0.74	7	400
40	20	primary	H2b	0.00	0.02	0	400
50	20	heavy-tailed	H1a	1.00	1.00	980933	200
50	20	heavy-tailed	H1b	0.54	0.84	13	200
50	20	heavy-tailed	H2a	0.42	0.68	6	200
50	20	heavy-tailed	H2b	0.00	0.01	0	200
50	20	lit-centred	H1a	1.00	1.00	98826	200
50	20	lit-centred	H1b	0.62	0.87	16	200
50	20	lit-centred	H2a	0.50	0.84	10	200
50	20	lit-centred	H2b	0.00	0.01	0	200
50	20	primary	H1a	1.00	1.00	988071	400
50	20	primary	H1b	0.58	0.82	14	400
50	20	primary	H2a	0.44	0.73	8	400
50	20	primary	H2b	0.00	0.01	0	400

Table 6

Probability of Exceeding BF > 10 (Primary) and BF > 3 (Secondary) per Design Condition (continued)

N	Verbs	Prior	Hyp	P(BF>10)	P(BF>3)	Median BF	Sims
50	20	weak	H1a	1.00	1.00	981423	200
50	20	weak	H1b	0.72	0.88	25	200
50	20	weak	H2a	0.37	0.69	6	200
50	20	weak	H2b	0.01	0.04	0	200
60	20	primary	H1a	1.00	1.00	999000	400
60	20	primary	H1b	0.72	0.90	27	400
60	20	primary	H2a	0.48	0.75	9	400
60	20	primary	H2b	0.00	0.01	0	400
70	20	primary	H1a	1.00	1.00	991535	200
70	20	primary	H1b	0.72	0.92	35	200
70	20	primary	H2a	0.56	0.78	13	200
70	20	primary	H2b	0.00	0.00	0	200
80	20	primary	H1a	1.00	1.00	999499	200
80	20	primary	H1b	0.81	0.96	45	200
80	20	primary	H2a	0.44	0.74	8	200
80	20	primary	H2b	0.00	0.01	0	200
90	20	primary	H1a	1.00	1.00	995260	200
90	20	primary	H1b	0.85	0.96	67	200
90	20	primary	H2a	0.58	0.82	16	200
90	20	primary	H2b	0.00	0.00	0	200
100	20	primary	H1a	1.00	1.00	1004258	200
100	20	primary	H1b	0.88	0.97	112	200
100	20	primary	H2a	0.57	0.76	15	200
100	20	primary	H2b	0.00	0.01	0	200

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